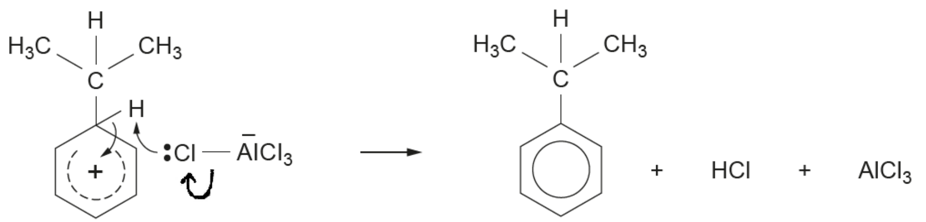
<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>9</td><td>(a)</td><td>(i)</td><td></td><td>(concentrated) nitric acid and (concentrated) sulfuric acid</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>tin/iron and (concentrated) hydrochloric acid</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>separation problems – the boiling temperature of the three isomers are too close together

accept other sensible answers</td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(iv)</td><td></td><td><table><tr><td>Reagent</td><td>$FeCl_3$</td><td>$NaHCO_3$</td></tr><tr><td>Observation</td><td>purple colour</td><td>no change</td></tr></table></td><td>1</td><td>1</td><td></td><td>2</td><td></td><td>2</td></tr>
<tr><td></td><td>(b)</td><td></td><td></td><td>$\rightarrow$ 5C + 2CO + N_2 + 3H_2O</td><td></td><td>1</td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td></td><td>

award (1) for curly arrows – must have arrow into benzene ring and one other
award (1) for all three products</td><td>1</td><td>1</td><td></td><td>2</td><td></td><td></td></tr>
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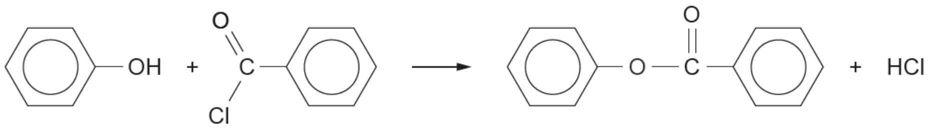
Question				**Marking details**	**Marks available**					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(d)	(i)		n(benzene) = $\frac{234 \times 1000}{78}$ = 3000 (1) n(phenol) at 86% yield = $\frac{3000 \times 86}{100}$ = 2580 mass of phenol = $\frac{2580 \times 94}{1000}$ = 243 kg (1)		1 1		2	1	
		(ii)		a species with an unpaired electron	1			1		
		(iii)		award (1) for any radical e.g. $\bullet CH_3$ $\bullet Cl$ $\bullet CH_2Cl$		1		1		1
	(e)			award (1) for either of following solution remains yellow / orange no more white precipitate is formed			1	1		
	(f)	(i)				1		1		
		(ii)		CH_3COCl will react (preferentially) with the <u>NaOH / water</u>			1	1		1
		(iii)		pyridine acts as a base / removes H^+ (1) as its nitrogen atom has a lone pair (of electrons) (1)			2	2		
				Question 9 total	**5**	**7**	**5**	**17**	**1**	**4**